ENGINEERING CHEMISTRY (UNIT - I)

Complete Concise Revision & Exam-Ready Theory Notes | B.Tech 1st Year (HBTU Kanpur)

Subject: Engineering Chemistry (B.Tech First Year) • **Syllabus Scope:** All 10 Core Topics **Format:** High-Yield Quick Review Booklet

1. Valence Shell Electron Pair Repulsion (VSEPR) Theory

Proposed by **Gillespie and Nyholm (1957)** to predict the 3D geometry and molecular shapes of polyatomic species based on minimizing electrostatic repulsions between valence electron pairs on the central atom.

Fundamental Postulates:

- The molecular geometry is governed by the total number of valence shell electron pairs (bond pairs, bp, and lone pairs, lp) around the central atom.
- Electron pairs occupy localized spatial orbitals and repel each other. They orient at maximum possible distances and angles to achieve minimum electrostatic potential energy (maximum stability).
- **Order of Repulsive Magnitude:**

  Lone Pair – Lone Pair ($lp - lp$) > Lone Pair – Bond Pair ($lp - bp$) > Bond Pair – Bond Pair ($bp - bp$)

  Physical Reason: Lone pair electrons are attracted by only one positive nucleus, having a broader, more diffused charge cloud. Bond pairs are held between two nuclei, making them spatially restricted.
- Multiple bonds ($C{=}C$, $C{\equiv}C$) are treated as single electron domains ("super-pairs"), but higher electron density causes greater repulsive force than single bonds.

Steric No.	Electron Geometry	bp	lp	Molecular Shape	Bond Angle	Standard Examples & Explanation
2	Linear	2	0	**Linear**	180°	$BeCl_2$, CO_2, HCN, C_2H_2
3	Trigonal Planar	3	0	**Trigonal Planar**	120°	BF_3, BCl_3, SO_3, NO_3^-
3	Trigonal Planar	2	1	**Bent / V-shaped**	< 120° (119.5°)	SO_2, O_3, NO_2^- (lp compresses O–S–O)
4	Tetrahedral	4	0	**Tetrahedral**	109.5° (109°28')	CH_4, CCl_4, SiF_4, NH_4^+, SO_4^{2-}
4	Tetrahedral	3	1	**Trigonal Pyramidal**	107°	NH_3, PCl_3, H_3O^+ (1 lp repels 3 bp)
4	Tetrahedral	2	2	**Bent / Angular**	104.5°	H_2O, H_2S, OF_2, SCl_2 (2 lp compress angle)
5	Trigonal Bipyramidal	5	0	**Trigonal Bipyramidal**	120° (eq), 90° (ax)	PCl_5 (axial bonds longer due to 3 × 90° repulsions)
5	Trigonal Bipyramidal	4	1	**See-Saw (Distorted Tet.)**	< 120°, < 90°	SF_4 (lone pair occupies equatorial site)
5	Trigonal Bipyramidal	3	2	**T-shaped**	~87.5° (< 90°)	ClF_3, BrF_3 (both lp reside in equatorial plane)
5	Trigonal Bipyramidal	2	3	**Linear**	180°	XeF_2, I_3^-, ICl_2^- (3 lp at 120° in equatorial belt)
6	Octahedral	6	0	**Octahedral**	90°	SF_6, $[PF_6]^-$, $[AlF_6]^{3-}$
6	Octahedral	5	1	**Square Pyramidal**	< 90° (~85°)	BrF_5, IF_5, $XeOF_4$
6	Octahedral	4	2	**Square Planar**	90°	XeF_4, $[ICl_4]^-$ (2 lp occupy opposite trans axial sites)
7	Pentagonal Bipyramidal	7	0	**Pentagonal Bipyramidal**	72° (eq), 90° (ax)	IF_7

> **Bent's Rule & Axial/Equatorial Site Selection:**
> *"More electronegative substituents prefer hybrid orbitals with less s-character, whereas lone pairs and electropositive groups prefer hybrid orbitals having higher s-character."* In sp^3d (TBP = $sp^2_{(eq)} + pd_{(ax)}$), lone pairs reside at equatorial positions (33.3% s-character) to minimize severe 90° repulsions. In octahedral sp^3d^2, lone pairs prefer opposite axial trans positions (180° apart) to minimize mutual repulsion.

2. Valence Bond Theory (VBT) & Coordination Chemistry

Pioneered by **Heitler and London (1927)** and developed by **Linus Pauling (1931)**. Describes covalent bonding as quantum-mechanical constructive interference and overlap of half-filled atomic orbitals with antiparallel electron spins.

Core Postulates & Types of Overlap:

- A covalent bond forms when two valence atomic orbitals overlap. Greater spatial overlap results in greater electron density between nuclei and a stronger, more stable bond.
- **Sigma (σ) Bond:** Formed by axial/head-on overlap along the internuclear axis ($s{-}s$, $s{-}p_x$, $p_x{-}p_x$). Cylindrically symmetric, higher bond strength, allows free rotation.
- **Pi (π) Bond:** Formed by lateral/sideways parallel overlap perpendicular to the bond axis ($p_y{-}p_y$, $p_z{-}p_z$). Electron density resides above and below the nodal plane; weaker than σ-bond; prevents free rotation.
- **Hybridization Concept:** Pauling introduced mathematical mixing of non-equivalent atomic orbitals of the same atom having similar energy to produce equal numbers of completely equivalent *hybrid orbitals* with identical energy, shape, and directional properties (sp, sp^2, sp^3, dsp^2, sp^3d, sp^3d^2, d^2sp^3).

Application of VBT to Coordination Complexes:

- The central transition metal ion provides empty valence hybrid orbitals equal to its coordination number (CN) to accept coordinate covalent lone pairs donated by ligands.
- **Inner Orbital Complexes (Low Spin):** Employs inner $(n-1)d$ orbitals in hybridization → d^2sp^3 (Octahedral). Strong field ligands (e.g., CN^-, CO, NO_2^-) force pairing of electrons in $(n-1)d$ subshell, leaving inner d orbitals vacant. These complexes are generally diamagnetic or possess low paramagnetism.
  Example: $[Fe(CN)_6]^{4-} \rightarrow Fe^{2+}\ (3d^6)$ pairs up into $3d_{xy}^2\ 3d_{yz}^2\ 3d_{xz}^2$; uses two inner $3d$, one $4s$, and three $4p$ → d^2sp^3 → Diamagnetic ($n = 0$, $\mu = 0$ BM).
- **Outer Orbital Complexes (High Spin):** Employs outer nd orbitals → sp^3d^2 (Octahedral). Weak field ligands (e.g., F^-, Cl^-, H_2O) cannot force pairing. Electrons remain unpaired according to Hund's rule; highly paramagnetic.
  Example: $[FeF_6]^{3-} \rightarrow Fe^{3+}\ (3d^5)$ has 5 unpaired electrons; uses outer $4s$, $4p^3$, $4d^2$ → sp^3d^2 → Paramagnetic ($n = 5$, $\mu = 5.92$ BM).
- **Magnetic Moment Formula:**

$$\mu_s = \sqrt{n(n+2)}\ BM$$ (where n = number of unpaired electrons; BM = Bohr Magneton, $1\ ext\{\ BM\} = eh/4\pi mc$)

> **Critical Limitations of Valence Bond Theory:**
> 1. Purely qualitative; does not explain why certain ligands force pairing (strong field) while others do not (weak field).
> 2. Cannot explain the electronic absorption spectra (origin of color) of transition metal complexes.
> 3. Fails to quantitatively predict magnetic susceptibilities and their variation with temperature.
> 4. Cannot explain tetrahedral vs. square planar geometry preference in 4-coordinate d^8 systems (e.g., $[Ni(CN)_4]^{2-}$ square planar vs. $[NiCl_4]^{2-}$ tetrahedral).
> 5. Completely ignores dynamic and static Jahn-Teller distortions.

3. Crystal-Field Theory (CFT) & d-Orbital Splitting

Formulated by **Hans Bethe (1929)** and **J.H. Van Vleck (1935)**. CFT assumes the metal-ligand interaction is **purely electrostatic (100% ionic)**. Ligands are treated as negative point charges (anions) or point dipoles with negative ends pointing at the central metal cation.

d-Orbital Splitting in an Octahedral (O_h) Field:

In an isolated gaseous ion, all five d-orbitals are degenerate. When 6 ligands approach along the Cartesian coordinate axes ($\pm x, \pm y, \pm z$):

- **e_g Orbitals ($d_{x^2-y^2}, d_{z^2}$):** Lobes point directly along the axes toward the incoming ligand charges → experience maximum electrostatic repulsion → raised in energy by **$+0.6\,\Delta_o$ (+6 Dq)** above the barycenter.
- **t_{2g} Orbitals (d_{xy}, d_{yz}, d_{xz}):** Lobes point between the axes (at 45° to axes) → experience significantly less repulsion → stabilized in energy by **$-0.4\,\Delta_o$ (–4 Dq)** below the barycenter.
- Total splitting energy: $\Delta_o = 10\ Dq$. Conservation of energy around the barycenter: $3(-0.4\,\Delta_o) + 2(+0.6\,\Delta_o) = 0$.

OCTAHEDRAL FIELD (O

$_h$) Free Ion (d^5) Spherical Field

e_g ($d_{x^2-y^2}$, d_{z^2}) +0.6 Δ_o (+6 Dq) t_{2g} (d_{xy}, d_{yz}, d_{xz}) -0.4 Δ_o (-4 Dq) Δ_o TETRAHEDRAL FIELD (T_d) t_2 (d_{xy}, d_{yz}, d_{xz}) +0.4 Δ_t (+4 Dq) e ($d_{x^2-y^2}$, d_{z^2}) -0.6 Δ_t (-6 Dq) Δ_t = 4/9 Δ_o

Figure: Comparison of Crystal Field Splitting in Octahedral (Oh) and Tetrahedral (Td) Geometries

Tetrahedral Splitting (T_d):

- In T_d geometry (CN = 4), ligands approach along directions that point between Cartesian axes (toward alternating corners of a cube).
- Non-axial t_2 orbitals lie closer to ligand directions → raised in energy by $+0.4\,\Delta_t$.
- Axial e orbitals lie further from ligand approach → lowered by $-0.6\,\Delta_t$. Note: Subscript 'g' is omitted because tetrahedral geometry lacks a center of inversion (i).
- **Splitting Relationship:**

$$\Delta_t = (4/9) \times \Delta_o \approx 0.44\,\Delta_o$$

Because 4 ligands exert fewer repulsions than 6 and none point directly at d-orbitals, Δ_t is always significantly smaller than pairing energy ($\Delta_t < P$). Consequently, **all tetrahedral complexes are High-Spin**.

Crystal Field Stabilization Energy (CFSE) Calculation:

CFSE is the thermodynamic energy stabilization resulting from the placement of electrons into split d-orbitals compared to an unsplit spherical field:

For Octahedral: $CFSE = [-0.4 \times n(t_{2g}) + 0.6 \times n(e_g)] \times \Delta_o + m \times P$

For Tetrahedral: $CFSE = [-0.6 \times n(e) + 0.4 \times n(t_2)] \times \Delta_t$

where $n(t_{2g}), n(e_g)$ are electron counts in respective levels, P is pairing energy, and m is extra electron pairs formed beyond free ion state.

Config.	High Spin (Weak Field, Δ_o < P)	CFSE (High Spin)	Low Spin (Strong Field, Δ_o > P)	CFSE (Low Spin)
d^1	$t_{2g}{}^{1}\,e_g{}^{0}$	$-0.4\ \Delta_o$	$t_{2g}{}^{1}\,e_g{}^{0}$	$-0.4\ \Delta_o$
d^2	$t_{2g}{}^{2}\,e_g{}^{0}$	$-0.8\ \Delta_o$	$t_{2g}{}^{2}\,e_g{}^{0}$	$-0.8\ \Delta_o$
d^3	$t_{2g}{}^{3}\,e_g{}^{0}$	$-1.2\ \Delta_o$	$t_{2g}{}^{3}\,e_g{}^{0}$	$-1.2\ \Delta_o$
d^4	$t_{2g}{}^{3}\,e_g{}^{1}$ (unpaired in e_g)	$-0.6\ \Delta_o$	$t_{2g}{}^{4}\,e_g{}^{0}$ (paired in t_{2g})	$-1.6\ \Delta_o + 1P$
d^5	$t_{2g}{}^{3}\,e_g{}^{2}$ (half-filled)	$0.0\ \Delta_o$ (Zero CFSE)	$t_{2g}{}^{5}\,e_g{}^{0}$	$-2.0\ \Delta_o + 2P$
d^6	$t_{2g}{}^{4}\,e_g{}^{2}$	$-0.4\ \Delta_o$	$t_{2g}{}^{6}\,e_g{}^{0}$ (fully filled t_{2g})	$-2.4\ \Delta_o + 2P$ (Maximum Low-spin stability)
d^7	$t_{2g}{}^{5}\,e_g{}^{2}$	$-0.8\ \Delta_o$	$t_{2g}{}^{6}\,e_g{}^{1}$	$-1.8\ \Delta_o + 1P$
$d^8 - d^{10}$	*Configurations d^8 ($t_{2g}{}^6\,e_g{}^2$), d^9 ($t_{2g}{}^6\,e_g{}^3$), and d^{10} ($t_{2g}{}^6\,e_g{}^4$) are identical in both High-Spin and Low-Spin fields.*			

4. Ligands (Strong & Weak Field) & Electronic Spectra

Strong Field Ligands (Δ_o > P):

- Possess large crystal field splitting power.
- Force electrons to pair in lower t_{2g} level before entering e_g → Form **Low-Spin** / **Inner Orbital** complexes.
- Typically strong π-acceptor ligands: $CO, CN^-, NO_2^-, en, dipy$.

Weak Field Ligands (Δ_o < P):

- Possess small crystal field splitting power.
- Electrons occupy higher e_g level before pairing in t_{2g} (Hund's rule holds) → Form **High-Spin** / **Outer Orbital** complexes.
- Typically π-donor / halide ligands: $I^-, Br^-, Cl^-, F^-, OH^-, H_2O$.

The Spectrochemical Series:

An experimentally determined sequence of ligands arranged in order of increasing crystal field splitting magnitude (Δ_o):

$I^- < Br^- < S^{2-} < SCN^- < Cl^- < NO_3^- < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < CH_3CN < NH_3 <$ en < bipy < phen $< NO_2^- < PPh_3 < CN^- <$ CO

Factors Affecting the Magnitude of Crystal Field Splitting (Δ):

- **Oxidation State of Metal Cation:** Higher positive charge draws ligands closer, exerting stronger electrostatic field → Δ for M^{3+} is ~40–50% greater than for M^{2+} (e.g., $[Fe(H_2O)_6]^{3+} > [Fe(H_2O)_6]^{2+}$).
- **Principal Quantum Number (3d vs. 4d vs. 5d):** Going down a transition group, d-orbitals become larger and more spatially diffused, overlapping more strongly with ligands → Δ increases by ~30–50% from $3d \rightarrow 4d \rightarrow 5d$. (Hence, 4d and 5d complexes are virtually all low-spin).
- **Geometry of Complex:** $\Delta_{sq.\ planar} > \Delta_{octahedral} > \Delta_{tetrahedral}$ ($\Delta_t = 4/9\,\Delta_o$; Δ_{sp} pprox $1.3\,\Delta_o$).

Origin of Electronic Spectra and Color:

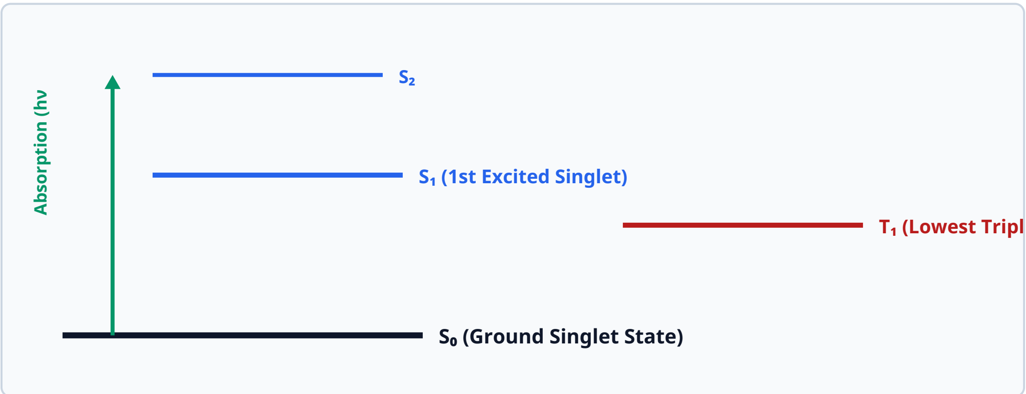
Transition metal complexes exhibit striking colors because their split d-orbitals have an energy gap ($\Delta E = h\nu = hc/\lambda$) matching the visible photon spectrum (400–750 nm). When a complex absorbs a specific wavelength, the human eye observes the **transmitted complementary color**.

- **d–d Transitions:** Promotion of an electron from lower t_{2g} to higher e_g.
  Classic Example: $[Ti(H_2O)_6]^{3+}$ (d^1) has a single electron in t_{2g} ($t_{2g}{}^1\,e_g{}^0$). It absorbs yellow-green light at *λ pprox 500 ext{ nm}* ($\bar{\nu}$ *= 20,000 ext{ cm}^{-1}*) to promote the electron to e_g ($t_{2g}{}^0\,e_g{}^1$). It transmits complementary red and blue light, appearing **violet**.

- **Charge-Transfer (CT) Spectra:** Transitions between metal and ligand molecular orbitals with very high molar absorptivity (ε_{max} $> 10,000$ ext{ L·mol}^{-1} ext{cm}^{-1}):
  - **LMCT (Ligand-to-Metal Charge Transfer):** Electron transfers from ligand-centered non-bonding orbital to empty metal d-orbitals. Explains intense colors in complexes with d^0 metal ions where d-d transitions are impossible: MnO_4^- (Mn^{7+}, d^0, deep purple), $Cr_2O_7^{2-}$ (Cr^{6+}, d^0, bright orange).
  - **MLCT (Metal-to-Ligand Charge Transfer):** Electron transfers from low-valent metal d-orbitals into empty π* antibonding orbitals of aromatic ligands (e.g., $[Fe(bipy)_3]^{2+}$, intense blood red).
- **Electronic Selection Rules:**
  - **Laporte Selection Rule (Parity Rule):** In centrosymmetric molecules, transitions between states of the same parity are forbidden ($g \to g$ or $u \to u$ forbidden; $g \leftrightarrow u$ allowed; $\Delta l = \pm 1$). Since all d-orbitals are 'gerade' (g), pure $d{-}d$ transitions are Laporte-forbidden. They gain weak intensity (ε pprox $1 - 100$) through vibronic coupling (loss of temporary center of inversion via molecular vibration). Non-centrosymmetric tetrahedral complexes are Laporte-free and have much stronger colors.
  - **Spin Selection Rule:** Transitions between states of different spin multiplicity are strictly forbidden ($\Delta S = 0$). Spin-forbidden transitions (e.g., high-spin d^5 $[Mn(H_2O)_6]^{2+}$) have extremely pale/faint color ($\varepsilon < 0.1$).

5. Photochemistry: Fluorescence, Phosphorescence & Jablonski Diagram

When an organic or inorganic molecule absorbs a quantum of ultraviolet or visible light ($h\nu$), an electron is excited from the ground singlet electronic state (S_0) to an upper excited singlet state (S_1, S_2). Relaxation back to the ground state proceeds via competing non-radiative and radiative processes, mapped by the **Jablonski Diagram**.



A ~10^{-15} s) IC (10^{-12} s)

Fluorescence ($h\nu_F$) $S_1 \to S_0$ (10^{-9} to 10^{-7} s) ISC (10^{-8} s) Spin Inversion Phosphorescence ($h\nu_P$) $T_1 \to S_0$ (10^{-4} to sec)

Figure: Complete Jablonski Diagram illustrating Absorption, Non-Radiative (IC, ISC), and Radiative (Fluorescence, Phosphorescence) Pathways

1. Non-Radiative Deactivation Processes (No Light Emitted):

- **Vibrational Relaxation (VR):** Hot excited molecules lose excess vibrational energy to surrounding solvent molecules via molecular collisions within 10^{-12} s.
- **Internal Conversion (IC):** Radiationless iso-energetic transition between electronic states of the *same spin multiplicity* ($S_2 \to S_1$ or $T_2 \to T_1$). Rapid (10^{-12} s).
- **Intersystem Crossing (ISC):** Radiationless crossover between electronic states of *different spin multiplicities* ($S_1 \to T_1$). Involves spin inversion; strictly spin-forbidden but promoted by strong spin-orbit coupling (presence of heavy atoms like Br, I, Pt). Timescale: 10^{-8} s.

2. Radiative Deactivation Processes (Light Emitted):

Characteristics	Fluorescence	Phosphorescence
Electronic Transition	$S_1 \rightarrow S_0$ (Singlet to Singlet ground state)	$T_1 \rightarrow S_0$ (Triplet to Singlet ground state)
Spin Multiplicity	Spin-allowed ($\Delta S = 0$); electron spin remains unchanged	Spin-forbidden ($\Delta S \neq 0$); requires quantum spin inversion
Lifetime (τ)	Extremely fast: 10^{-9} *ext{ to }* 10^{-6} seconds (nanoseconds)	Long-lived / delayed: 10^{-3} seconds to minutes or hours
Emission Behavior	Stops immediately (< 10^{-8} s) when exciting light source is removed	Persistent emission continues (delayed "afterglow" in dark)
Wavelength Order	$\lambda_{abs} < \lambda_{fluorescence} < \lambda_{phosphorescence}$	Longer wavelength (lower energy) than fluorescence because $T_1 < S_1$ (Hund's Rule)
Key Applications	Fluorescein, Optical brighteners in detergents, DNA tagging, bio-imaging	Glow-in-the-dark emergency signage, watch dials, security printing inks

Key Photophysical Laws:
- **Stokes Shift:** The difference in position between absorption maximum and emission maximum ($\lambda_{emission} > \lambda_{absorption}$). Caused by fast vibrational energy dissipation in the excited state prior to emission.
- **Kasha's Rule:** Photon emission (fluorescence or phosphorescence) occurs with appreciable yield only from the lowest excited electronic state (S_1 or T_1).
- **Quantum Yield (Φ):** Ratio of photons emitted to photons absorbed: Φ = (*ext{No. of molecules emitting photons}) / (ext{No. of quanta absorbed})*.

6. Hydrogen Bonding & Effect on Physical Properties

A specialized dipole-dipole electrostatic attraction formed between a covalently bonded hydrogen atom attached to a strongly electronegative, small atom (F, O, N) and an unshared lone pair of electrons on an adjacent electronegative atom. Bond energy typically ranges from **10 to 40 kJ/mol** (weaker than true covalent bonds, but ~10× stronger than van der Waals forces).

1. Intermolecular Hydrogen Bonding:
- Forms between two or more independent molecules of the same or different substances.
- Causes **molecular association** into oligomeric clusters.
- *Examples:* H_2O, HF, NH_3, alcohols ($R-OH$), acetic acid dimer ($(CH_3COOH)_2$), p-nitrophenol.

2. Intramolecular Hydrogen Bonding (Chelation):
- Forms internally within the same individual molecule between two proximate functional groups.
- Forms a stable 5- or 6-membered planar chelate ring; *prevents* intermolecular association.
- *Examples:* o-Nitrophenol, Salicylaldehyde, Salicylic acid.

Pronounced Effects on Physical Properties:

- **Boiling Point and Melting Point:**
  - Intermolecular H-bonding requires substantial thermal energy to overcome associated molecular networks, leading to abnormally high boiling points.
    Classic Case: H_2O (BP = 100°C) is a liquid at ambient temperature due to extensive 3D H-bonding networks, while heavier group 16 analogues (H_2S BP = –60°C, H_2Se BP = –41°C) are gases because sulfur lacks sufficient electronegativity to form strong H-bonds.
  - **Comparison of Isomers (o-Nitrophenol vs. p-Nitrophenol):**
    - *o-Nitrophenol:* Undergoes **intramolecular H-bonding** → molecule is self-contained and non-associated → lower boiling point (216°C), highly steam-volatile, lower water solubility.
    - *p-Nitrophenol:* Undergoes extensive **intermolecular H-bonding** → molecules associate into chains → higher boiling point (279°C), non-steam volatile, higher water solubility. (Allows easy separation via steam distillation).
- **Solubility in Polar Solvents:** Covalent organic substances that can establish H-bonds with water molecules dissolve readily (e.g., lower alcohols, glucose, acetone), whereas hydrocarbons, alkyl halides, and benzene cannot form H-bonds and remain insoluble.
- **Viscosity and Surface Tension:** Extensive intermolecular H-bonding impedes fluid layers from sliding past each other:

Viscosity Trend: Glycerol ($CH_2OH{-}CHOH{-}CH_2OH$, 3 –OH groups) > Ethylene Glycol (2 –OH) > Ethanol (1 –OH) > Diethyl Ether (0 –OH)

- **Density Anomaly and Open-Cage Structure of Ice:** In crystalline ice, each oxygen atom is tetrahedrally bonded to four hydrogen atoms (two by covalent bonds at ~1.0 Å, two by H-bonds at ~1.76 Å). This creates a rigid, open, hexagonal cage-like structure containing large empty interstitial voids. Upon heating from 0°C to 4°C, thermal vibrations break some H-bonds, the open cage collapses, and water molecules pack closer together. Hence, **liquid water is denser than ice (ice floats), reaching maximum density at 3.98°C (~4°C)**.

7. Metallic Bonding: Theories & Characteristic Properties

Metallic bonding is the strong, non-directional electrostatic cohesive force binding positive metal ions (kernels) to a surrounding delocalized electron cloud.

1. Free Electron Sea Model (Lorentz-Drude Theory):

- Metals have low ionization energies and vacant valence orbitals. Metal atoms shed their valence electrons, becoming fixed positive cations (called **kernels**) organized in a close-packed crystal lattice.
- The liberated valence electrons are not bound to any individual nucleus but constitute a mobile, highly delocalized **"sea of electrons"** permeating the interstitial spaces of the entire lattice.
- **Microscopic Explanations of Metallic Properties:**
  - *Electrical Conductivity:* In the absence of an electric field, free electrons move randomly. Under an applied electric potential gradient, electrons experience a directional drift toward the positive terminal, establishing a steady current.
  - *Thermal Conductivity:* Heat applied at one end increases the kinetic vibrational energy of local electrons. Mobile electrons rapidly move and collide with adjacent particles, transporting thermal energy rapidly across the metal.
  - *Metallic Luster:* Incident light photons cause oscillatory vibrations of surface mobile electrons. The accelerating electrons instantly re-emit electromagnetic radiation of the same frequencies, conferring a shiny, reflective metallic luster.
  - *Malleability & Ductility:* Because metallic bonding is completely non-directional, mechanical shear forces cause entire planes of metal kernels to slide past one another without shattering. The flexible electron sea instantly flows to cushion the displaced kernels and preserve cohesive bonding.

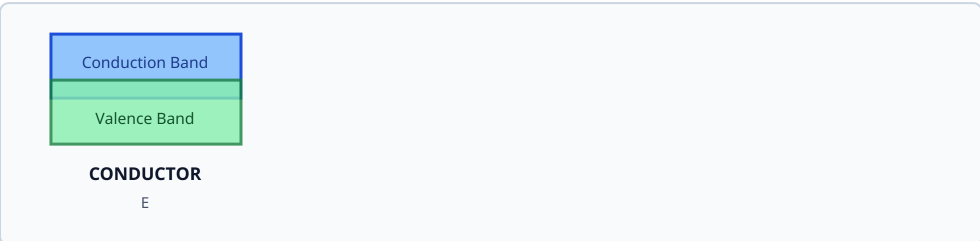
2. Molecular Orbital / Band Model Approach:

According to Molecular Orbital Theory, when N identical metal atoms combine, their valence atomic orbitals overlap to generate N discrete molecular orbitals. Because N is on the order of Avogadro's number (*pprox 10^{23}*), the energy separation between adjacent orbital levels is exceedingly tiny ($\sim 10^{-20}$ *ext{ eV}*), coalescing into continuous **energy bands**.

8. Band Theory of Solids & Superconductors

Classification of Solids via Band Theory:

Solids are categorized into three distinct regimes depending on the magnitude of the **Forbidden Energy Gap (E_g)** between the highest filled **Valence Band (VB)** and the lowest empty **Conduction Band (CB)**:



g = 0 (Overlapping) CB (Empty at 0 K)

$E_g \leq 1.5$ eV VB (Filled) SEMICONDUCTOR CB (Completely Empty) $E_g > 5$ eV VB (Completely Full) INSULATOR

Figure: Energy Band Diagrams depicting Forbidden Gap (Eg) in Conductors, Semiconductors, and Insulators

Class	Band Gap ( )	Conductivity (, S·cm^{-1})	Temperature Effect & Charge Carriers
Conductors	$E_g = 0$ (Overlapping VB and CB or half-filled band)	Very High (10^4 ext{ to } 10^7)	Conductivity *decreases* as temperature rises (R increases due to enhanced lattice thermal vibrations/phonon scattering). Free electrons are carriers.
Semiconductors	Small gap ($0.2 < E_g \leq 2.0$ ext{ eV}; Si = 1.1 eV, Ge = 0.67 eV)	Moderate (10^{-6} ext{ to } 10^4)	Conductivity *increases exponentially* with temperature (electrons thermally promoted from VB → CB, leaving mobile holes in VB). Intrinsic and Extrinsic (n-type: P/As donor; p-type: B/Al acceptor).
Insulators	Very large forbidden gap ($E_g > 3 - 6$ ext{ eV}; Diamond = 5.47 eV)	Negligible ($< 10^{-10}$)	Thermal energy at room temperature is completely incapable of exciting electrons across the wide forbidden gap. No current flows.

Superconductivity:

Discovered by **Heike Kamerlingh Onnes (1911)** in liquid mercury at 4.2 K. Superconductivity is a macroscopic quantum phenomenon in which certain materials lose all electrical resistance below a characteristic **Critical Temperature (T_c)**.

- **Zero Electrical Resistance ($R = 0$):** A current injected into a closed superconducting ring flows perpetually without potential drop or ohmic power dissipation.
- **Meissner Effect:** When cooled below T_c inside an external magnetic field (H), the superconductor spontaneously expels all magnetic flux lines from its bulk interior ($B = 0$). The material acts as a **perfect diamagnet** with volume magnetic susceptibility $\chi = -1$ ($B = \mu_0(H + M) = 0 \rightarrow M = -H$).
- **BCS Theory (Bardeen, Cooper, Schrieffer, 1957):** Electrons near the Fermi level pair up into **Cooper pairs** mediated by virtual phonon (lattice vibration) interactions. The resulting paired bosons condense into a single coherent macroscopic quantum ground state that traverses the crystal lattice without scattering.

Property	Type-I Superconductors (Soft)	Type-II Superconductors (Hard)
Critical Fields	Single critical magnetic field (H_c)	Two critical magnetic fields: Lower (H_{c1}) and Upper (H_{c2})
Transition State	Abrupt, sharp transition from superconducting to normal state at H_c	Exhibits an intermediate **vortex** / **mixed state** between H_{c1} and H_{c2} where magnetic flux penetrates in quantized vortex lines
Meissner Effect	Strictly obeys Meissner effect up to H_c	Complete Meissner expulsion up to H_{c1}; partial flux penetration between H_{c1} and H_{c2}
Critical Field Values	Low ($H_c \leq 0.1$ ext{ Tesla}); easily destroyed by self-induced fields	Extremely high ($H_{c2} > 30$ ext{ to } 100 ext{ Tesla})
Materials & Usage	Pure metals: Pb, Hg, Sn, Al. (Unusable for high-field electromagnets).	Alloys / Ceramics: $NbTi, Nb_3Sn$, High-T_c Cuprates like YBCO ($YBa_2Cu_3O_{7-\delta}$, T_c pprox 93 ext{ K} > boiling point of liquid nitrogen 77 K). Used in MRI, Maglev trains, and LHC particle accelerators.

9. Liquid Crystals (Mesomorphic State): Classification & Applications

Discovered by Austrian botanist **Friedrich Reinitzer (1888)**. Liquid crystals represent a thermodynamically stable intermediate state of matter (a **mesophase**) situated between an anisotropic crystalline solid (possessing 3D positional and orientational order) and an isotropic liquid (possessing neither).

Primary Classification:

- **Thermotropic Liquid Crystals:** Mesophase transitions are driven purely by temperature changes. Typically formed by elongated, rod-like (calamitic) or disc-like (discotic) organic molecules with rigid aromatic cores and flexible alkyl tails.
- **Lyotropic Liquid Crystals:** Formed when amphiphilic molecules (containing both hydrophilic polar heads and hydrophobic hydrocarbon tails) are dissolved in an isotropic solvent (e.g., water). Phase transitions depend on concentration and temperature (e.g., soaps, detergents, phospholipid cell bilayers).

Classification of Thermotropic Mesophases:

Mesophase	Molecular Arrangement & Structural Features	Typical Examples & Characteristics
1. Nematic Phase (Greek *nema* = thread)	Molecules maintain **long-range orientational order** (long axes point roughly parallel along a preferred direction described by the **director** $\hat{n}$), but have **no positional order**. Molecules can translate and slide freely in all three dimensions. Highly fluid; displays characteristic thread-like schlieren textures under polarizing microscopes.	**MBBA** (4-methoxybenzylidene-4'-butylaniline), **5CB** (4-cyano-4'-pentylbiphenyl). Primary active component in LCDs due to low viscosity and fast dielectric response.
2. Smectic Phase (Greek *smegma* = soap)	Molecules possess both **orientational order and 1D positional order** arranged in discrete, well-defined 2D parallel layers: • *Smectic A (SmA):* Molecular director is perpendicular (normal) to layer planes. • *Smectic C (SmC):* Molecular director is tilted at a specific angle relative to layer normal. Much more viscous than nematics.	**Ethyl p-azoxybenzoate**, Octylcyanobiphenyl (8CB). Used in high-resolution optical storage media and bistable memory displays.
3. Cholesteric Phase (Chiral Nematic, N^*)	Formed by chiral molecules. Structurally equivalent to nematic layers stacked on top of one another, but the director turns continuously by a small azimuth angle from layer to layer, tracing out a macroscopic **helical twist**. The pitch (p) is the distance along the helix axis for a full 360° director rotation. Reflects circularly polarized light matching Bragg condition: $\lambda = n_{avg} \times p$.	**Cholesteryl nonanoate**, cholesteryl benzoate. The pitch is exquisitely sensitive to minute temperature changes, producing brilliant, reversible color shifts across the visible spectrum.
4. Discotic Phase	Molecules consist of a flat, disc-shaped rigid aromatic core with flexible aliphatic chains radiating outward. Discs stack face-to-face into columnar 1D wires that arrange in 2D hexagonal lattices.	Triphenylene and hexasubstituted benzene derivatives. Applied in organic one-dimensional conductors, field-effect transistors, and organic photovoltaics.

Technological & Industrial Applications of Liquid Crystals:

1. Liquid Crystal Displays (LCDs): Exploits Twisted Nematic (TN) and In-Plane Switching (IPS) electro-optical effects. In a TN cell, nematic liquid crystal molecules are anchored in a 90° helical twist between perpendicular polarizers. Applying a small electric potential unwinds the molecules parallel to the electric field, eliminating light rotation and turning the pixel black (smartphones, computer monitors, TVs).
2. Medical Thermography & Clinical Diagnosis: Cholesteric liquid crystal mixtures painted on skin undergo color changes corresponding to surface temperature gradients, visually highlighting subcutaneous vascular anomalies, localized inflammation, vein thrombosis, and early-stage malignant breast tumors.
3. Non-Destructive Testing (NDT) in Engineering: Coated on printed circuit boards (PCBs), aerospace composites, and microelectronic chips to identify hot-spots, short circuits, microscopic hairline fractures, and stress concentrations.
4. Smart Windows & Optical Devices: Polymer Dispersed Liquid Crystals (PDLC) switch rapidly from an opaque, scattering translucent state (no voltage) to a clear transparent state upon applying voltage.

10. Spectroscopy: UV-Visible & Infrared (IR)

Spectroscopy studies the interaction of electromagnetic radiation with matter. Photon absorption promotes transitions between quantized molecular energy levels (electronic, vibrational, or rotational), enabling structural identification and quantitative assay.

Part A: UV-Visible Spectroscopy (Electronic Transitions)

- **Wavelength Spectrum:** Ultraviolet (200–400 nm) and Visible (400–800 nm). Vacuum UV (< 200 nm) requires evacuated chambers due to atmospheric oxygen absorption.
- **Fundamental Principle:** Absorption of UV-Vis photons imparts sufficient energy ($\sim 100 - 600$ ext{ kJ/mol}) to promote electrons from occupied bonding or non-bonding molecular orbitals to unoccupied antibonding molecular orbitals.
- **Types of Electronic Transitions (in order of decreasing excitation energy ΔE):**

$$\sigma \rightarrow \sigma^* > n \rightarrow \sigma^* > \pi \rightarrow \pi^* > n \rightarrow \pi^*$$

  - $\sigma \rightarrow \sigma^*$: Requires very high energy vacuum UV (< 180 nm). Exhibited by saturated hydrocarbons (CH_4, C_2H_6) containing only single C–C and C–H σ-bonds.

- $n \rightarrow \sigma^*$: Non-bonding lone pair electron promoted to σ* orbital (180–220 nm). Observed in saturated compounds containing heteroatoms (CH_3OH, CH_3Cl, CH_3NH_2).
- $\pi \rightarrow \pi^*$: Excitation of π-bonding electron to π* antibonding orbital (200–400 nm). Characteristic of unsaturated bonds ($C{=}C$, $C{\equiv}C$, alkenes, alkynes, aromatics). Shows intense absorption ($\varepsilon_{max} > 10{,}000$). Conjugation decreases the HOMO-LUMO gap and shifts absorption to longer wavelengths.
- $n \rightarrow \pi^*$: Non-bonding electron promoted to π* orbital (270–320 nm). Observed in unsaturated compounds with heteroatoms ($C{=}O, C{=}S, -NO_2$). Symmetry-forbidden transition → low molar absorptivity (ε_{max} pprox $10 - 100$).

- **Governing Law: Beer-Lambert Law:**

$$A = \log_{10}(I_0 / I) = \varepsilon \times c \times l$$

where A = Absorbance (optical density, dimensionless), I_0 = incident light intensity, I = transmitted intensity, ε = molar absorptivity / extinction coefficient ($\text{L·mol}^{-1} \text{cm}^{-1}$), c = concentration (mol/L), and l = path length (cm).

- **Chromophores & Auxochromes:**
  - **Chromophore:** An isolated covalently unsaturated functional group responsible for light absorption in the UV-Vis region (e.g., $-C{=}C-, -C{\equiv}C-, -C{=}O, -NO_2, -N{=}N-$).
  - **Auxochrome:** A saturated group containing non-bonding electron pairs that does not absorb UV radiation on its own, but when attached directly to a chromophore, enhances absorption intensity and shifts absorption to longer wavelengths (e.g., $-OH, -NH_2, -NR_2, -OCH_3, -SH, -Cl$).

Spectral Shift	Common Name	Direction of Shift	Underlying Chemical Cause
Bathochromic Shift	Red Shift	Shift of λ_{max} to **longer wavelength** (lower energy)	Conjugation extension, attachment of electron-donating auxochrome, solvent polarity.
Hypsochromic Shift	Blue Shift	Shift of λ_{max} to **shorter wavelength** (higher energy)	Removal of conjugation, protonation of auxochrome (e.g., aniline → anilinium ion).
Hyperchromic Effect	Intensity Increase	Increase in molar absorptivity ($\varepsilon_{max} \uparrow$)	Introduction of auxochrome, structural planarization.
Hypochromic Effect	Intensity Decrease	Decrease in molar absorptivity ($\varepsilon_{max} \downarrow$)	Steric hindrance twisting molecular planarity, distortion of π-conjugation.

Instrumentation of Double-Beam UV-Vis Spectrophotometer:

- **Light Sources:** Deuterium (D_2) lamp emits continuous spectrum across UV range (190–380 nm); Tungsten-Halogen lamp covers visible and near-IR range (380–900 nm).
- **Monochromator:** High-precision holographic diffraction grating with entrance/exit slits to isolate monochromatic wavelength bands ($\pm 1 \text{ nm}$).
- **Rotating Beam Chopper:** Alternating sector mirror that splits the monochromatic beam into two equivalent paths: sample beam and reference beam (100–200 Hz).
- **Sample & Reference Cuvettes:** Precision rectangular optical cells of 1.0 cm path length fabricated from **Fused Quartz / Silica**. (Standard glass cannot be used because it absorbs UV radiation below 350 nm).
- **Detector & Readout:** Photomultiplier Tube (PMT) or Silicon Photodiode array converts transmitted photons into electrical currents, and a differential amplifier computes $\log(I_0/I)$.

Part B: Infrared (IR) Spectroscopy (Vibrational Transitions)

- **Spectral Range:** Analytical mid-IR spans wavenumbers $4000 \text{ to } 400 \text{ cm}^{-1}$ ($\lambda = 2.5 \text{ to } 25 \text{ }\mu\text{m}$). Wavenumber $\bar{\nu} = 1/\lambda = \nu/c$.
- **Condition for IR Activity (Selection Rule):**

A molecular vibration is IR Active if and only if it produces a periodic change in the molecular dipole moment:

$$d\mu/dx \neq 0$$

  - *IR Active:* Heteronuclear molecules with dynamic dipole moments (HCl, CO, H_2O, SO_2).

- *IR Inactive:* Homonuclear diatomic molecules (H_2, N_2, O_2, Cl_2) have zero dipole moment in all vibrational states. In linear CO_2, symmetric stretching maintains inversion symmetry ($\mu = 0$) and is IR inactive, while asymmetric stretching and bending modes alter the dipole and are IR active.

- **Fundamental Modes of Vibration (Degrees of Freedom):**

  Non-Linear Molecules: Total fundamental vibrations = $3N - 6$ (e.g., H_2O: $3(3)-6 = 3$ modes)
  Linear Molecules: Total fundamental vibrations = $3N - 5$ (e.g., CO_2: $3(3)-5 = 4$ modes)

  - **Stretching Vibrations (changes bond length):** Symmetric stretching (v_s) and Asymmetric stretching (v_{as}).
  - **Bending / Deformational Vibrations (changes bond angle):** In-plane bending (Scissoring, Rocking) and Out-of-plane bending (Wagging, Twisting). *Note:* Stretching requires greater restorative force than bending; hence stretching peaks appear at higher wavenumbers.

- **Hooke's Law for Vibrational Frequency:**
  Treating a chemical bond as a harmonic oscillator with two vibrating atomic masses m_1 and m_2:

$$\bar{v} = (1 / 2\pi c) \times \sqrt{(k / \mu)} \quad (\text{in cm}^{-1})$$

where k = bond force constant (k_{triple} pprox 15 imes 10^5 ext{ dyn/cm} > k_{double} pprox 10 imes 10^5 > k_{single} pprox 5 imes 10^5), and $\mu = (m_1 m_2)/(m_1+m_2)$ is the reduced mass.
Deductions: Stronger bonds vibrate at higher frequencies ($C{\equiv}C > C{=}C > C{-}C$); lighter vibrating atoms vibrate at higher frequencies ($C{-}H > C{-}C > C{-}Cl$).

- **Spectral Regions: Functional Group vs. Fingerprint Region:**
  - **Functional Group Region (*4000 − 1400 ext{ cm}^{-1}*):** Characteristic stretching frequencies of specific functional groups appear here, almost independent of the remainder of the molecular skeleton.
  - **Fingerprint Region (*1400 − 400 ext{ cm}^{-1}*):** Arises from complex single-bond deformations and skeletal bending modes ($C{-}C$, $C{-}O$, $C{-}N$). Highly idiosyncratic to each specific molecule; identical to a human fingerprint for compound verification.

Functional Group & Mode	Characteristic Range (cm^{-1})	Spectral Features & Analytical Utility
O–H (Alcohol / Phenol)	*3200 − 3600 ext{ cm}^{-1}*	Very broad and strong band due to intermolecular H-bonding. Free non-bonded O–H gives a sharp band at ~*3650 ext{ cm}^{-1}*.
O–H (Carboxylic Acid)	*2500 − 3300 ext{ cm}^{-1}*	Extremely broad, jagged band centered at ~3000 cm^{-1}, overlapping aliphatic C–H stretching.
N–H (Amines / Amides)	*3300 − 3500 ext{ cm}^{-1}*	Primary amine ($-NH_2$): characteristic doublet (2 peaks). Secondary amine ($-NH-$): single sharp peak. Tertiary: no band.
C–H Stretch (Aliphatic / Aromatic)	*2850 − 3300 ext{ cm}^{-1}*	sp^3 C–H (< 3000 cm^{-1}); sp^2 =C–H / aromatic (> 3000–3100 cm^{-1}); sp ≡C–H (sharp at 3300 cm^{-1}).
C≡C and C≡N (Triple Bonds)	*2100 − 2260 ext{ cm}^{-1}*	Sharp, medium-to-strong band. $C{\equiv}N$ nitrile at 2250 cm^{-1}; alkyne $C{\equiv}C$ at 2150 cm^{-1}.
C=O (Carbonyl Group)	*1670 − 1750 ext{ cm}^{-1}*	Intensely strong, sharp peak. Aldehydes (1725), Saturated ketones (1715), Esters (1735), Acids (1710), Amides (1680 cm^{-1}).
C=C (Alkene / Aromatic Ring)	*1600 − 1680 ext{ cm}^{-1}*	Alkene at 1650 cm^{-1}. Benzene ring displays sharp doublet/multiplet at 1600 and 1475 cm^{-1}.

Instrumentation & Sample Preparation in IR:

- **IR Sources:** Nernst Glower (zirconium, yttrium, and thorium oxides heated electrically to 1500°C) or Globar (silicon carbide rod heated to 1300°C).
- **Optical Windows:** Glass and quartz cannot be used because they absorb strongly in the mid-IR; windows and prisms are made of alkali halide crystals such as **NaCl or KBr**.

- **Sample Handling:**
  - *Solids:* **KBr Pellet Method** (sample pulverized with optical-grade KBr and pressed under 10-ton hydraulic die into transparent discs) or **Nujol Mull** (ground with mineral paraffin oil).
  - *Liquids:* Examined as a thin film squeezed between two polished NaCl plates. Water is strictly forbidden as solvent because it dissolves NaCl plates and strongly absorbs IR.

Summary Comparison: UV-Vis vs. IR Spectroscopy

• **UV-Vis Spectroscopy:** Involves valence electronic excitations; requires chromophores and π-conjugation; governed by Beer-Lambert Law; primarily used for **quantitative concentration determination** and extent of conjugation.

• **IR Spectroscopy:** Involves molecular vibrational excitations; requires net dynamic change in dipole moment ($d\mu/dx \neq 0$); governed by Hooke's Law; primarily used for **qualitative functional group identification** and structural fingerprinting.

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